Shiv Nadar University-Chennai

Kalavakkam, Tamil Nadu - 603110, India

Website: www.snuchennai.edu.in

Department of Computer Science and Engineering

CAPSTONE PROJECT REPORT
Phase 1 - Review 3

Tiny Object Detection in Aerial Images

Submitted by

Student Name	Registration Number
Anjana K	23011101022
Lalith Adithyan S	23011101071

Under the guidance of

Dr. Chandrakala S

Professor

Approved By:

Dr. Chandrakala S
Project Guide

Chapter 3

Dataset / Simulation Details

3.1 Dataset or Simulation Environment

The evaluation of the object detection models is conducted on two primary large-scale aerial benchmarks: the VisDrone2019 dataset and the DOTA-v1.0 dataset. These datasets were explicitly chosen due to their rigorous representation of real-world aerial perception challenges. UAV imagery is characterized by extreme scale variation, ranging from massive infrastructures to minute pedestrians, where small objects are easily overwhelmed by complex backgrounds. Furthermore, non-uniform object distributions and varying drone flight altitudes introduce severe spatial heterogeneity. In this project, the primary focus is directed toward detecting highly dense object clusters and camouflaged targets that naturally blend into their surrounding environments, making them exceptionally difficult to isolate from background noise.

3.2 Dataset Description

The project relies on two distinct aerial datasets to validate both low-altitude and high-altitude detection robustness against dense and visually obscured targets:

1. VisDrone2019 Dataset

This dataset consists of images captured by various drone platforms across multiple urban and rural environments under different weather and lighting conditions. It comprises 6,471 training images, 548 validation images, and 1,610 testing images[cite: 104]. The dataset features 10 target categories: pedestrian, person, car, van, bus, truck, motor, bicycle, awning-tricycle, and tricycle[cite: 104].



Figure 3.1: Sample images from the VisDrone2019 dataset showcasing high-density small objects in urban environments.

2. DOTA-v1.0 Dataset

DOTA-v1.0 is a large-scale dataset for object detection in aerial images, containing 2,806 images with spatial dimensions ranging from 800×800 to 4000×4000 pixels[cite: 23]. It contains 188,282 annotated object instances across 15 categories: plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter, soccer ball field, roundabout, and swimming pool[cite: 23].

3.3 Data Collection and Preprocessing

Due to the massive resolution and varied formats of the source imagery, a rigorous preprocessing pipeline is applied before feeding the data into the detection network:

- **Image Tiling and Cropping:** Because DOTA-v1.0 images can reach up to 4000×4000 pixels, feeding them directly into the network exceeds GPU memory limits. The source images are cropped into 1024×1024 patches

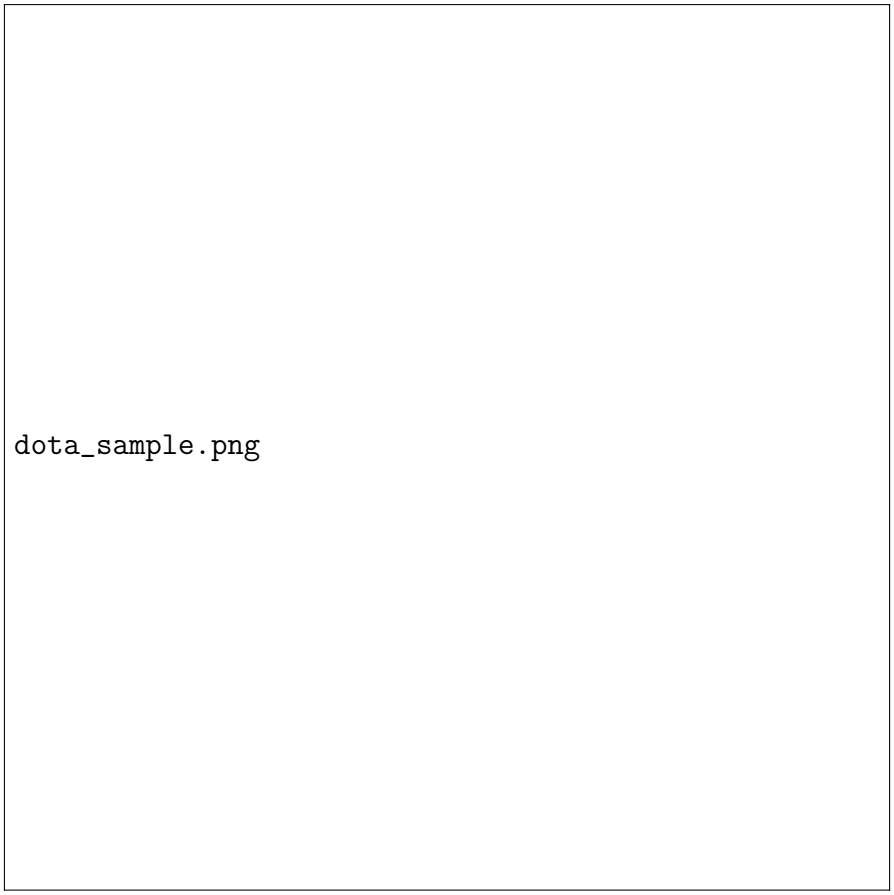


Figure 3.2: Sample aerial images from the DOTA-v1.0 dataset demonstrating extreme scale variations and arbitrary object orientations.

with a 200-pixel overlap to ensure targets on the boundaries are not severed. VisDrone images are resized to a standardized 640×640 resolution.

- **Spatio-Frequency Extraction:** To address the camouflage effect and textural confusion (e.g., trash resembling cars), preprocessing includes the extraction of a Laplacian of Gaussian (LoG) cue ($\sigma \in \{1, 2, 4\}$). This process separates high-frequency structural edges from low-frequency background noise, creating a vital control signal that will guide the network's scale-sequence scanning.

- **Data Augmentation:** To prevent overfitting and improve the model's robustness against complex background clutter, several augmentation techniques are applied. These include Mosaic and MixUp to simulate heavily crowded scenes, alongside standard random flipping, random cropping, and HSV color space adjustments.

3.4 Dataset Characteristics

The defining characteristics of both datasets are the extreme scale imbalance, dense occlusion, and the presence of objects that camouflage into the terrain. In aerial photography, small targets occupy only a few pixels and exhibit extremely low contrast against complex backgrounds, effectively blending them into their surroundings. For instance, in the VisDrone dataset, small objects (sub-32 pixels) dominate the overall target distribution, while large objects represent a very small fraction of the data.

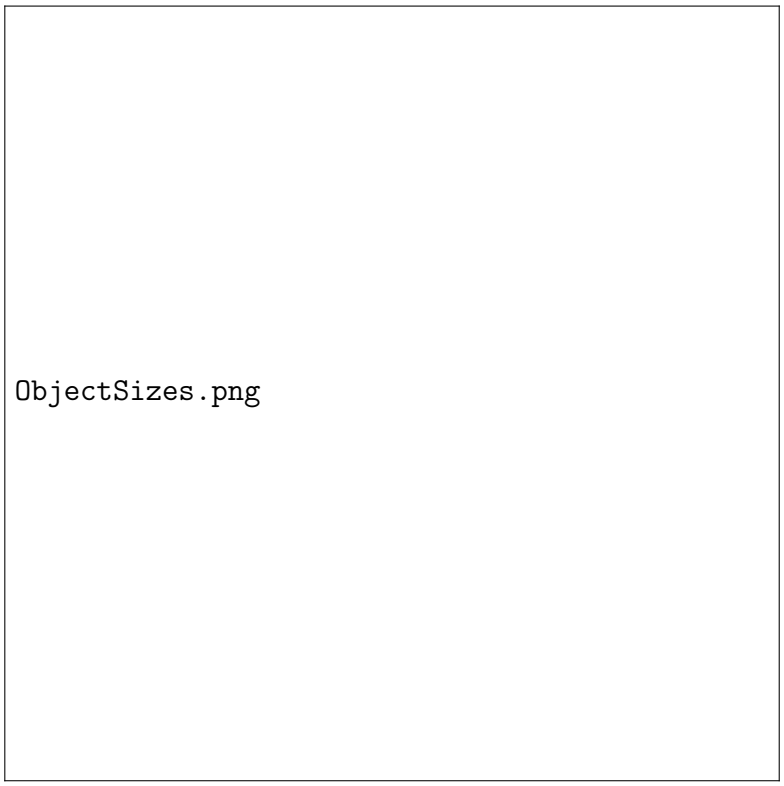


Figure 3.3: Proportion of small, medium, and large object instances, illustrating the heavy skew toward tiny targets in aerial imagery.

3.5 Assumptions and Limitations

While preparing the datasets for model evaluation, several constraints and assumptions were acknowledged:

1. **Optical Clarity Assumption:** The primary datasets assume relatively clear optical conditions. While varying illumination is present, severe atmospheric degradations (such as dense fog, heavy rain, or complete darkness) are underrepresented. Detecting highly camouflaged or super tiny objects in

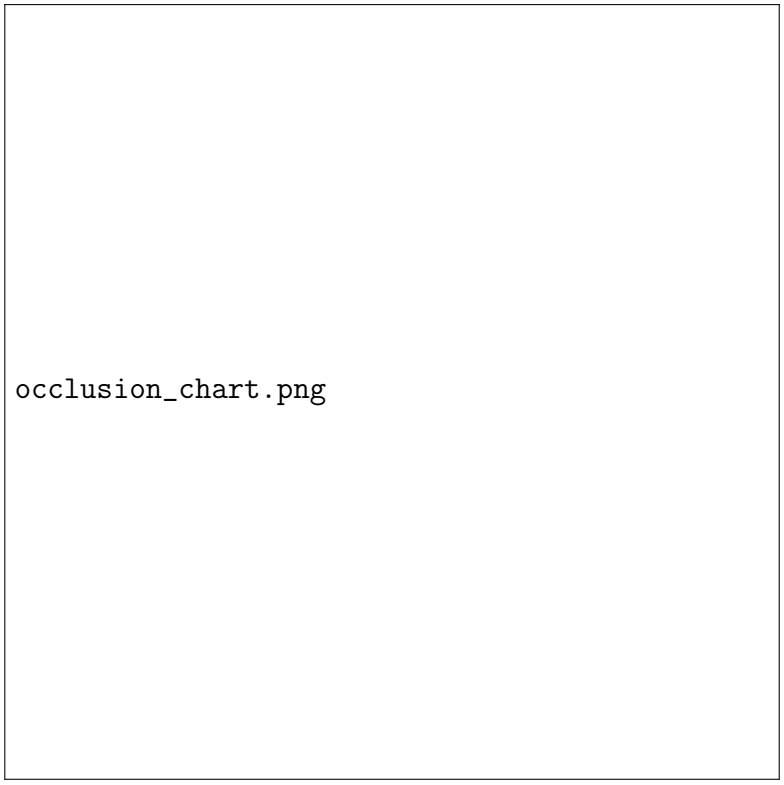


Figure 3.4: Occlusion distribution per category, highlighting the severe target overlapping present in drone-captured scenes.

unseen haze remains a fundamental limitation without specialized domain adaptation.

2. **Annotation Noise in Dense Clusters:** In ultra-dense scenes (e.g., massive crowds or congested parking lots), human annotation naturally introduces marginal localization errors or missed labels for highly occluded sub-16-pixel targets, which establishes a slight theoretical ceiling on supervised training precision.

3. **Hardware Constraints:** It is assumed that raw high-resolution aerial images cannot be processed in a single forward pass during real-time edge deployment. The necessity of sliding-window cropping or resizing inherently risks losing the absolute widest macro-contextual cues needed to distinguish certain camouflaged objects from the background.

Chapter 4

System Model

4.1 System Overview

Describe the proposed system and its major components. Explain how the system addresses the identified problem.

4.2 System Architecture

Present the overall architecture using a suitable block diagram. Explain the role of each component and its interactions.

4.3 System Workflow

Describe the step-by-step operation of the proposed system. Include a flowchart or sequence diagram wherever appropriate.